

PROBLEM SET 1:

```
alma15@alma15-VirtualBox:~/Desktop/IBA_SEM5_OS/xv6-labs-2026$ echo "Name: Alma Mashhood | ERP: 30529"
Name: Alma Mashhood | ERP: 30529
alma15@alma15-VirtualBox:~/Desktop/IBA_SEM5_OS/xv6-labs-2026$ make qemu
mv -f fs.img fs.img.bk
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sync user/_sleep user/_sixfive user/_find user/_mempdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1155 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ █
```

SLEEP:

```
alma15@alma15-VirtualBox:~/Desktop/IBA_SEM5_OS/xv6-labs-2026$ echo "Name: Alma Mashhood | ERP: 30529"
Name: Alma Mashhood | ERP: 30529
alma15@alma15-VirtualBox:~/Desktop/IBA_SEM5_OS/xv6-labs-2026$ make qemu
mv -f fs.img fs.img.bk
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sync user/_sleep user/_sixfive user/_find user/_mempdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1155 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ sleep 10
$ █
```

SIXFIVE:

```
alma15@alma15-VirtualBox:~/Desktop/IBA_SEM5_OS/xv6-labs-2026$ echo "Name: Alma
Mashhood | ERP: 30529"
Name: Alma Mashhood | ERP: 30529
alma15@alma15-VirtualBox:~/Desktop/IBA_SEM5_OS/xv6-labs-2026$ make qemu
mv -f fs.img fs.img.bk
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo
user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir
user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_z
ombie user/_logstress user/_forphan user/_dorphan user/_sync user/_sleep user/
_sixfive user/_find user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks
1953 total 2000
balloc: first 1155 blocks have been allocated
balloc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -sm
p 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=no
ne,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ sixfive
10 15 7 30 42
10
15
30
42
$
```

MEMDUMP:

```
Name: Alma Mashhood | ERP: 30529
alma15@alma15-VirtualBox:~/Desktop/IBA_SEM5_OS/xv6-labs-2026$ make qemu
mv -f fs.img fs.img.bk
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_l
n user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forpha
n user/_dorphan user/_sync user/_sleep user/_sixfive user/_find user/_mendum
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1155 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -d
rive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ memdump
Example 1:
61810
2026
Example 2:
a string
Example 3:
another
Example 4:
D30
1819438967
100
z
xyzzzy
Example 5:
hello
w
o
r
l
d
$
```

FIND:

```
alma15@alma15-VirtualBox:~/Desktop/IBA_SEM5_OS/xv6-labs-2026$ echo "Name: Alma Mashhood
| ERP: 30529"
Name: Alma Mashhood | ERP: 30529
alma15@alma15-VirtualBox:~/Desktop/IBA_SEM5_OS/xv6-labs-2026$ make qemu
mv -f fs.img fs.img.bk
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_for
ktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh u
ser/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_fo
rphan user/_dorphan user/_sync user/_sleep user/_sixfive user/_find user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 tota
l 2000
balloc: first 1155 blocks have been allocated
balloc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nogra
phic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0
-device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ find . README
exec find failed
$ find . README
./README
$
```

EXEC:

```
almai5@almai5-VirtualBox:~/Desktop/IBA_SEMS_05/xv6-labs-2026$ echo "Name: Alma Mashhood | ERP: 30529"
Name: Alma Mashhood | ERP: 30529
almai5@almai5-VirtualBox:~/Desktop/IBA_SEMS_05/xv6-labs-2026$ make qemu
mv -f fs.ing fs.ing.bk
mkfs/mkfs fs.ing README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_
rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sync user/_sleep user/_sixfive user/
_find user/_mendum
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1155 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.ing,if=none,form
at=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ echo hello
hello
$ ls
.          1 1 1024
..         1 1 1024
README    2 2 2441
findtest.sh 2 3 91
sixfive.txt 2 4 19
cat        2 5 36728
echo       2 6 35592
forktest   2 7 18136
grep       2 8 44184
init       2 9 35976
kill       2 10 35512
ln         2 11 35304
ls         2 12 42880
mkdir      2 13 35568
rm         2 14 35552
sh         2 15 58312
stressfs   2 16 36416
usertests  2 17 205880
.          2 18 51824
grind      2 19 37728
wc         2 20 34848
zombie     2 21 37560
logstress  2 22 36352
forphan    2 23 35800
dorphan    2 24 34944
sync       2 25 35384
sleep      2 26 42496
sixfive    2 27 45416
find       2 28 44480
mendum     3 29 0
$
```

ASSIGNMENT 1:

UPTIME:

```
alma15@alma15-VirtualBox:~/Desktop/IBA_SEM5_05/xv6-labs-2026$ echo "Name: Alma Mashhood
| ERP: 30529"
Name: Alma Mashhood | ERP: 30529
alma15@alma15-VirtualBox:~/Desktop/IBA_SEM5_05/xv6-labs-2026$ make qemu
mv -f fs.img fs.img.bk
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_for
ktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh u
ser/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_fo
rphan user/_dorphan user/_sync user/_sleep user/_sixfive user/_find user/_uptime user/_r
awtest user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 tota
l 2000
ballocc: first 1244 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nogra
phic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0
-device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ uptime
52
$
```

RESEX:

```
alma15@alma15-VirtualBox:~/Desktop/IBA_SEM5_OS/xv6-labs-2026$ echo "Name: Alma Mashhood
| ERP: 30529"
Name: Alma Mashhood | ERP: 30529
alma15@alma15-VirtualBox:~/Desktop/IBA_SEM5_OS/xv6-labs-2026$ make qemu
mv -f fs.img fs.img.bk
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_for
ktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh u
ser/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_fo
rphan user/_dorphan user/_sync user/_sleep user/_sixfive user/_find user/_uptime user/_r
awtest user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 tota
l 2000
ballocc: first 1244 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nogra
phic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0
-device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ find . ^s
./sixfive.txt
./sh
./stressfs
./sync
./sleep
./sixfive
$
```

SHELL IMPROVEMENT:

#1:

```
alma15@alma15-VirtualBox:~/Desktop/IBA_SEMS_OS/xv6-labs-2026$ echo "Name: Alma Mashhood
| ERP: 30529"
Name: Alma Mashhood | ERP: 30529
alma15@alma15-VirtualBox:~/Desktop/IBA_SEMS_OS/xv6-labs-2026$ make qemu
mv -f fs.img fs.img.bk
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_for
ktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh u
ser/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_fo
rphan user/_dorphan user/_sync user/_sleep user/_sixfive user/_find user/_uptime user/_r
awtest user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 tota
l 2000
ballocc: first 1244 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nogra
phic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0
-device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ sh < findtest.sh
hello
hello
hello
$
```

#2:

```
alma15@alma15-VirtualBox:~/Desktop/IBA_SEMS_OS/xv6-labs-2026$ echo "Name: Alma Mashhood
| ERP: 30529"
Name: Alma Mashhood | ERP: 30529
alma15@alma15-VirtualBox:~/Desktop/IBA_SEMS_OS/xv6-labs-2026$ make qemu
mv -f fs.img fs.img.bk
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_for
ktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh u
ser/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_fo
rphan user/_dorphan user/_sync user/_sleep user/_sixfive user/_find user/_uptime user/_r
awtest user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 tota
l 2000
ballocc: first 1244 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nogra
phic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0
-device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ sleep 20 &
$ wait
$
```

#3:

```
alma15@alma15-VirtualBox:~/Desktop/IBA_SEM5_OS/xv6-labs-2026$ echo "Name: Alma Mashhood
| ERP: 30529"
Name: Alma Mashhood | ERP: 30529
alma15@alma15-VirtualBox:~/Desktop/IBA_SEM5_OS/xv6-labs-2026$ make qemu
mv -f fs.img fs.img.bk
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_for
ktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh u
ser/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_fo
rphan user/_dorphan user/_sync user/_sleep user/_sixfive user/_find user/_uptime user/_r
awtest user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 tota
l 2000
balloc: first 1244 blocks have been allocated
balloc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nogra
phic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0
-device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ sixf
```

AFTER PRESSING TAB AND ENTER:

```
alma15@alma15-VirtualBox:~/Desktop/IBA_SEMS_OS/xv6-labs-2026$ echo "Name: Alma Mashhood  
| ERP: 30529"  
Name: Alma Mashhood | ERP: 30529  
alma15@alma15-VirtualBox:~/Desktop/IBA_SEMS_OS/xv6-labs-2026$ make qemu  
mv -f fs.img fs.img.bk  
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_for  
ktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh u  
ser/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_fo  
rphan user/_dorphan user/_sync user/_sleep user/_sixfive user/_find user/_uptime user/_r  
awtest user/_memdump  
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 tota  
l 2000  
ballocc: first 1244 blocks have been allocated  
ballocc: write bitmap block at sector 46  
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nogra  
phic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0  
-device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0  
  
xv6 kernel is booting  
  
hart 2 starting  
hart 1 starting  
init: starting sh  
$ sixfive  
10 7 39 42  
10  
42
```

#4:

```
almai15@almai15-VirtualBox:~/Desktop/IBA_SEM5_05/xv6-labs-2026$ echo "Name: Alna Mashhood | ERP: 30529"
Name: Alna Mashhood | ERP: 30529
almai15@almai15-VirtualBox:~/Desktop/IBA_SEM5_05/xv6-labs-2026$ make qemu
mv -f fs.img fs.img.bk
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_m
kdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sync user/_sle
ep user/_sixfive user/_find user/_uptime user/_rawtest user/_mempdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1244 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,i
f=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0
```

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh

\$ echo hello

hello

\$ ls

.	1 1 1024
..	1 1 1024
README	2 2 2441
findtest.sh	2 3 91
sixfive.txt	2 4 19
cat	2 5 36776
echo	2 6 35640
forktest	2 7 18184
grep	2 8 44232
init	2 9 36024
kill	2 10 35560
ln	2 11 35352
ls	2 12 42928
mkdir	2 13 35616
rm	2 14 35600
sh	2 15 72536

rm	2 14 35600
sh	2 15 72536
stressfs	2 16 36472
usertests	2 17 205928
grind	2 18 51872
wc	2 19 37776
zombie	2 20 34896
logstress	2 21 37608
forphan	2 22 36400
dorphan	2 23 35848
sync	2 24 34992
sleep	2 25 35432
sixfive	2 26 42544
find	2 27 47816
uptime	2 28 35136
rawtest	2 29 35768
mempdump	2 30 44528
console	3 31 0

\$

AFTER PRESSING UP KEY AND ENTER:

```
rawtest      2 29 35768
memdump      2 30 44528
console      3 31 0
$ echo hello
hello
$ ls
.             1 1 1024
..            1 1 1024
README       2 2 2441
findtest.sh  2 3 91
sixfive.txt  2 4 19
cat           2 5 36776
echo         2 6 35640
forktest     2 7 18184
grep         2 8 44232
init         2 9 36024
kill         2 10 35560
ln           2 11 35352
ls           2 12 42928
mkdir        2 13 35616
rm           2 14 35600
sh           2 15 72536
stressfs     2 16 36472
usertests    2 17 205928
grind        2 18 51872
wc           2 19 37776
zombie       2 20 34896
logstress    2 21 37608
forphan      2 22 36400
dorphan      2 23 35848
sync         2 24 34992
sleep        2 25 35432
sixfive      2 26 42544
find         2 27 47816
uptime       2 28 35136
rawtest      2 29 35768
memdump      2 30 44528
console      3 31 0
$
```